

STAT3003: Survey Methods

midterm review

Main Contents

- 3 survey methods: SRS, Stratified random sampling, cluster sampling (and two-stage cluster sampling)
- 3 parameters: total τ , mean μ and proportion p
- 4 quantities: $E(\hat{\theta})$, $\widehat{Var}(\hat{\theta})$, $C.I.$, n

Basic Probability Rules

- conditional probability:

$$\Pr(A|B) = \frac{\Pr(A \text{ and } B)}{\Pr(B)}, \text{ thus } \Pr(A \text{ and } B) = \Pr(A|B) \Pr(B) = \Pr(B|A) \Pr(A)$$

- independence: $\Pr(A|B) = \Pr(A)$

- *Complement Rule: $P(A^c) = 1 - P(A)$

- *Addition Rule: if A and B are disjoint events,

$$\text{then } P(A \text{ or } B) = P(A) + P(B)$$

- *Full Probability Rule: $P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$

- *Multiplication Rule: if events A and B are independent, then $P(A \text{ and } B) = P(A) * P(B)$

- *Bayes' Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

Expectation/Mean and Variance

- Expected value

1. Discrete RV

$$E(X) \equiv \mu = \sum_{i=1}^k x_i \Pr(X = x_i)$$

2. Continuous RV

$$E(X) \equiv \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

■ Variance

$$\begin{aligned} Var(X) \equiv \sigma^2 &= E(X - \mu)^2 = \\ &= \sum_{i=1}^n (x_i - \mu)^2 \Pr(X = x_i) = E(X^2) - \mu^2 \end{aligned}$$

$$\begin{aligned} Var(X) \equiv \sigma^2 &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \\ &= \int_{-\infty}^{\infty} x^2 f(x) dx = E(X^2) - \mu^2 \end{aligned}$$

Binomial Distribution

$$P(X) = \frac{n!}{X! (n - X)!} p^X (1-p)^{n-X}$$

$$\mu = E(x) = np$$

$$\sigma^2 = np(1-p)$$

Special case: $n=1 \Rightarrow$ Bernoulli Distribution

Normal Distribution

- Probability density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty$$

for some parameters μ , σ , where $\sigma > 0$.

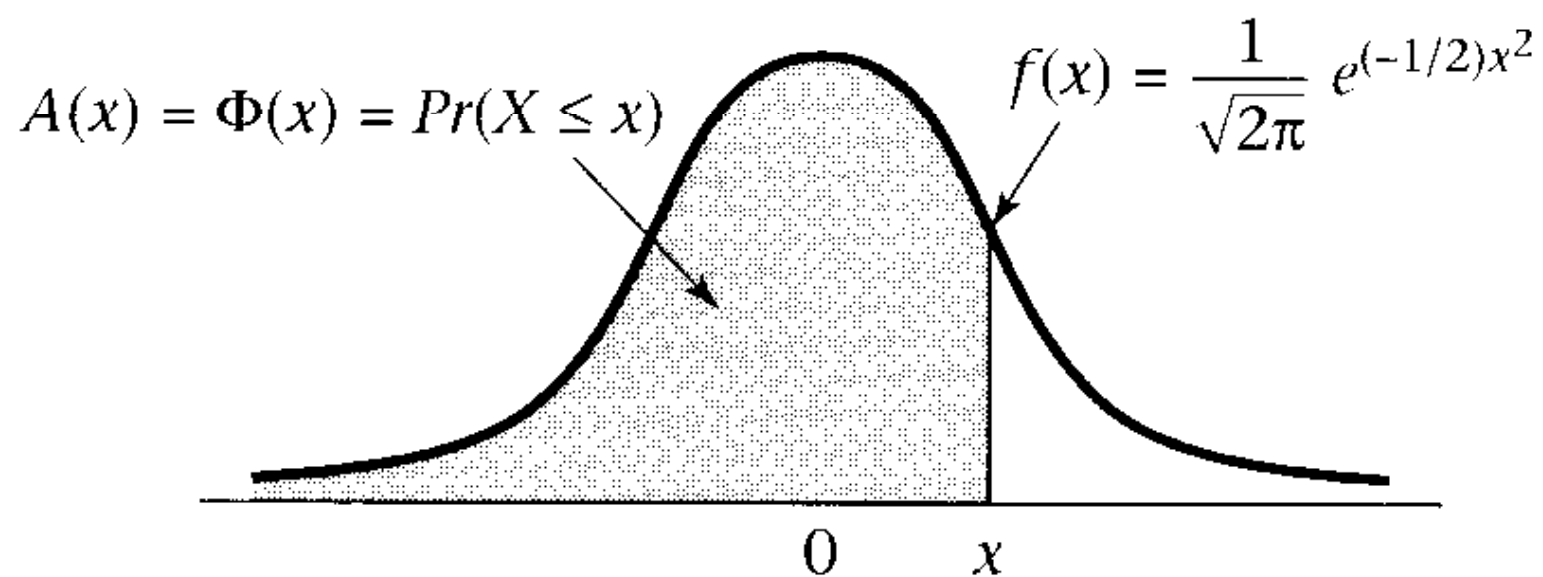
It turns out: $E(X)=\mu$, $Var(X)=\sigma^2$

- A special case: Standard Normal Distribution,
 $\mu=0$, $\sigma^2=1$

Symmetric about 0 since $f(x)=f(-x)$

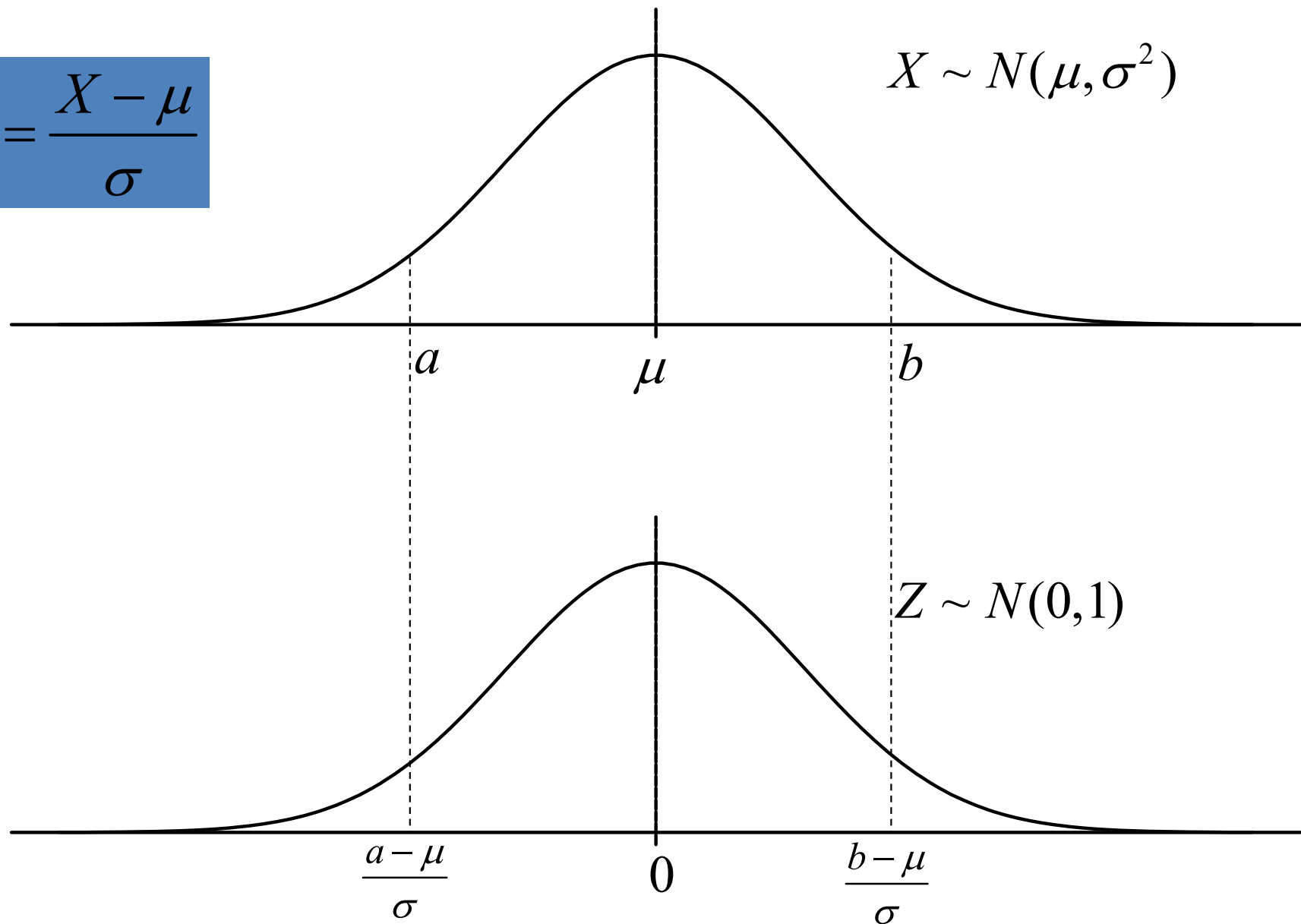
Use Tables for Standard Normal Distribution

Table 3 The normal distribution



(a)

$$Z = \frac{X - \mu}{\sigma}$$



Transformation from $N(\mu, \sigma^2)$ to $N(0, 1)$

We want $\Pr(a \leq X \leq b)$

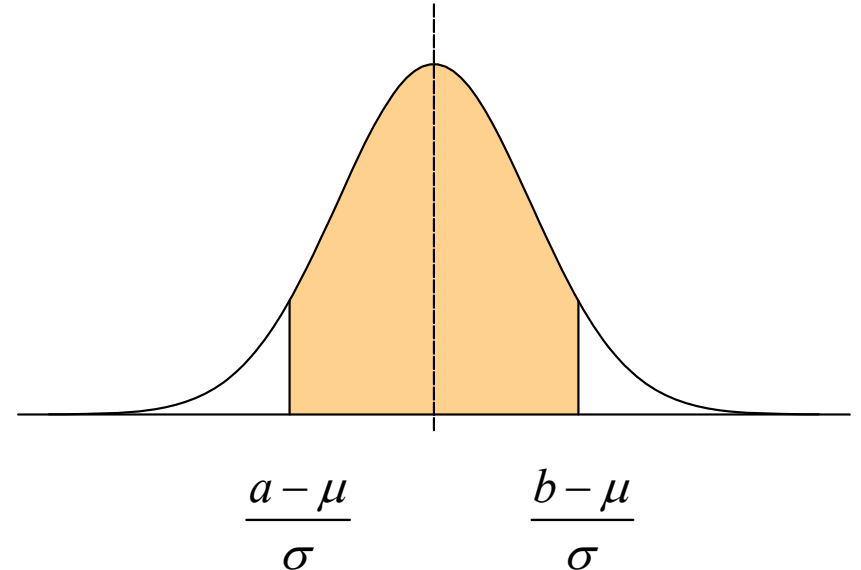
$X \sim N(\mu, \sigma^2)$, $b \geq a$

$$a \leq X \leq b$$

$$a - \mu \leq X - \mu \leq b - \mu$$

$$\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}$$

$$\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}$$



So,

$$\Pr(a \leq X \leq b) = \Pr\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)$$

$$= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

mean rule and variance rule

$$\mu_{aX+bY+c} = a\mu_X + b\mu_Y + c$$

$$Var(aX + bY + c) = \sigma_{aX+bY+c}^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y$$

population correlation coefficient: ρ
($\rho=0$ if independent)

Stronger result for Normal distribution:

if $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$

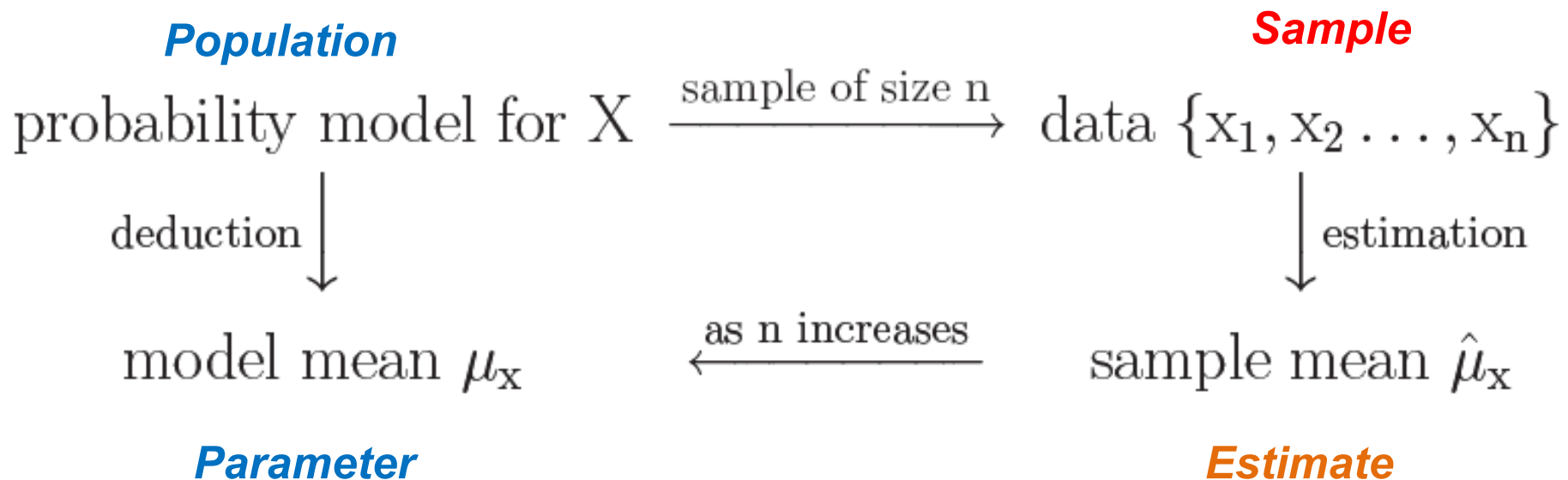
then $aX + bY + c \sim N(a\mu_X + b\mu_Y + c, a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y)$

Further if X & Y *are independent*

then $aX + bY + c \sim N(a\mu_X + b\mu_Y + c, a^2\sigma_X^2 + b^2\sigma_Y^2)$

Role in Statistics

- A **variable** is a characteristic measured on individuals/elements drawn from a population under study.
- **Data** are measurements of one or more variables made on a collection of individuals.
- Variable/Data Type: discrete, continuous, categorical



Estimation Accuracy

- $$\begin{aligned} E \left[(\hat{\theta} - \theta)^2 \right] &= E \left[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2 \right] \\ &= E \left[(\hat{\theta} - E[\hat{\theta}])^2 \right] + 2(E[\hat{\theta}] - \theta)E \left[\hat{\theta} - E[\hat{\theta}] \right] \\ &\quad + (E[\hat{\theta}] - \theta)^2 \\ &\Rightarrow \textcolor{red}{MSE}(\hat{\theta}) = \textcolor{red}{Var}(\hat{\theta}) + \textcolor{red}{Bias}(\hat{\theta})^2 \end{aligned}$$
- Therefore, **if estimator $\hat{\theta}$ is unbiased** (most of our estimators will be unbiased), then its **MSE equals its variance**

Estimation Accuracy

- Here are three useful words to describe an estimator
 - Unbiased if $E[\hat{\theta}] = \theta$
 - **Precise** if $Var(\hat{\theta})$ is small
 - **Accurate** if $E[\hat{\theta}] = \theta$ and $E[(\hat{\theta} - \theta)^2]$ is small
- Thus a precise estimator may not be accurate because it is biased.
- Precision considers how close estimates from different samples are to each other.
- Accuracy considers how close estimates are to the true value.

Population Parameters

- Population Mean

- Is the **average value** of all measurements in the population, i.e. $\frac{u_1 + u_2 + \dots + u_N}{N} := \mu$

- We will often denote the population mean by μ

Population Parameters

- **Finite Population Variance**

- The finite population variance is given by

- $$\frac{1}{N} \sum_{i=1}^N (u_i - \mu)^2 := \sigma^2$$

- We often denote the population variance by σ^2

- $\sigma^2 = \text{Var}(Y) = E[(Y - \mu)^2]$

- The **population standard deviation** (SD) simply the square root of the population variance

- $$\sqrt{\sum_{i=1}^N \frac{(u_i - \mu)^2}{N}} := \sigma$$

Population Parameters

- **Covariance**

- We have a finite population of size N . Each person in the population has two variables of interest, Y and X
- Let the population means of Y and X are μ_Y and μ_X respectively. Then the population covariance of X and Y is given by
 - $\frac{1}{N} \sum_{i=1}^N [(y_i - \mu_Y)(x_i - \mu_X)] := \text{Cov}(X, Y)$
- If X and Y are independent, then $\text{Cov}(X, Y)$ is zero
- The converse is not necessarily true
- It is tricky to decide whether a covariance is large or small because the scales of y and x are combined
- It would be better to have a dimensionless number to describe dependence...

Population Parameters

- Correlation

- Is that dimensionless quantity
- The correlation between X and Y is just the covariance of X and Y divided by the standard deviations of X and Y
- That is, $\frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y} := \rho$
- This must lie between -1 and 1 and it has no units
- We often denote it by ρ (rho)

Survey Terminology

- Example: I want to infer the proportion of Hong Kong high school students who support the use of ChatGPT for homework
- I decide I will ask 200 students
- I have a list of all the high schools in HK. I select 25 of these
- For each of the 25, I collect a name list from the school and I select 8 to interview
- **Element** in this survey is a Hong Kong high school student - we will “measure” them, i.e. record their response to a question.
- **Population** is all the high school students in Hong Kong.
- **Sampling units** partition the population (all the high school students in Hong Kong). We have two levels of sampling units in this survey - the high schools in Hong Kong; the individual high school students in Hong Kong.
- **Frame** is a list of sampling units. The first frame the survey uses is a list of all the high schools in Hong Kong - we randomly select 25 from this list. Then, for each of the 25 chosen schools, we have a list of high school students in that school - these are another 25 frames.
- We have two levels of **samples** in this survey: a sample of 25 out of the M many high schools in Hong Kong; a sample of 200 of the N many high school students in Hong Kong.

Quota Sampling

- Why use probability samples? Why not just choose elements carefully to ensure a **representative** sample?
- Samples which select elements to ensure the sample has **very similar characteristics** as the population are called **quota samples**
 - E.g. select a sample which has the same proportions of sex, age, income, education etc. as the population

Quota Sampling

- Problems
 - Instructions to interviewers in a quota sampling poll tended to be like “Find 2 men and 3 women in your block. Make sure 4 are over 25 years old and one is under 25.”
 - Instructions like these leave lots of room for human **subjectivity**. We want **objectivity**.
 - Perhaps the interviewer subconsciously (or consciously!) seeks out people who look “nicer” – e.g. richer and if Republicans tend to have higher incomes then...

Improve quota sampling?

- Maybe the quota sampling wasn't specific enough
 - Why not include more characteristics, like income level, marital status, attitude towards issues like taxes, healthcare, foreign policy, etc.
 - Or combinations of the above, e.g. match the ratio of white, single males with union membership
- Clearly it is impossible to control all factors which may influence a person's preference of Dewey or Truman
 - Indeed, trying to fix one ratio might disturb other ratios
- Randomisation balances out all factors inherent in a population
 - including the factors you know but would like to control
 - those you know but can't control
 - and those you don't know you don't know

Simple Random Sampling

- Simple Random Sampling (SRS) is
 - Random sampling **without replacement** such that every possible sample of n units is **equally likely**
- Therefore, there are $\frac{N!}{n!(N-n)!}$ possible random samples and each occurs with the same probability

Important tool: inclusion indicator Z

- Given a population of size N , what is the probability the i th element is in a SRS of size n ?
 - This is called the **inclusion probability** of element i
 - Let $\pi_i = P(\text{Element } i \text{ is in the sample})$
 - Then write **$\pi_i = E[Z_i]$** where Z_i is a random variable which equals 1 if element i is in the sample or 0 otherwise
 - **$E[Z_i] = \frac{n}{N}$**

How to do Simple Random Sampling

- 1st Method: Draw Lots
- 2nd Method: Use a Random Numbers Table
- 3rd Method: Use Random Numbers in Excel/R...

SRS: Properties of $\bar{Y}$

- Our estimator for μ is what we expect: **the sample mean**
- Let the population have values $\{u_1, u_2, \dots, u_N\}$
- Let $\{Y_1, Y_2, \dots, Y_n\}$ be the sample drawn by SRS without replacement
- The sample mean is given $\bar{Y} := \frac{1}{n} \sum_{i=1}^n Y_i$
 - We may write this another way: $\bar{Y} = \frac{1}{n} \sum_{j=1}^N u_j Z_j$

SRS: Properties of $\bar{Y}$

$$\text{➤ } E(\bar{Y}) = \frac{1}{n} \sum_{j=1}^N u_j \frac{n}{N} = \frac{1}{N} \sum_{j=1}^N u_j = \mu \text{ thus unbiased}$$

SRS: Properties of $\bar{Y}$

- Before we calculate $Var(\bar{Y})$, let us recall a useful result
 - Let $X_1, \dots, X_n$ be random variables and $a_1, \dots, a_n$ be constants
 - Then for $U := \sum_{i=1}^n a_i X_i$, we have
 - $Var(U) = \sum_{i=1}^n a_i^2 Var(X_i) + \sum_{i=1}^n \sum_{j \neq i} a_i a_j Cov(X_i, X_j)$
- Therefore, according to our useful result
 - $Var(\bar{Y}) = \sum_{j=1}^N \left(\frac{u_j}{n}\right)^2 Var(Z_j) + \sum_{j=1}^N \sum_{k \neq j} \frac{u_j}{n} \frac{u_k}{n} Cov(Z_j, Z_k)$
- Since Z_j is a Bernoulli random variable with probability of “success” $\frac{n}{N}$
 - $Var(Z_j) = \frac{n}{N} \left(1 - \frac{n}{N}\right)$

SRS: Properties of $\bar{Y}$

- Note

- $$Cov_{\left(\frac{n}{N}\right)^2}(Z_j, Z_k) = E(Z_j Z_k) - E(Z_j)E(Z_k) = E(Z_j Z_k) -$$

- Note $Z_j Z_k$ equals 0 unless both elements j and k are in the sample, then $Z_j Z_k = 1$

- The number of possible samples containing both j and k is $\frac{(N-2)!}{(n-2)!(N-n)!}$

- Each SRS sample is equally likely to occur, with the number of possible random samples $\frac{N!}{n!(N-n)!}$

- Hence $E(Z_j Z_k) = \frac{(N-2)!}{(n-2)!(N-n)!} \times \frac{1}{\frac{N!}{n!(N-n)!}} = \frac{n(n-1)}{N(N-1)}$

SRS: Properties of $\bar{Y}$

- Therefore

- $Cov(Z_j, Z_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)} = -\frac{n(1-\frac{n}{N})}{N(N-1)}$

- Thus

$$Var(\bar{Y}) = \frac{n}{N} \left(1 - \frac{n}{N}\right) \sum_{j=1}^N \frac{u_j^2}{n^2} - \frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{1}{N-1} \sum_{j=1}^N \sum_{k \neq j}^N \frac{u_j}{n} \frac{u_k}{n}$$
$$\Rightarrow Var(\bar{Y}) = \frac{1}{n^2} \frac{n}{N} \left(1 - \frac{n}{N}\right) \left[\sum_{j=1}^N u_j^2 - \frac{1}{N-1} \sum_{j=1}^N \sum_{k \neq j}^N u_j u_k \right]$$

Properties of $\bar{Y}$

- Now use the **identity**
 - $\sum_{j=1}^N (u_j - \mu)^2 = \frac{1}{N} [(N-1) \sum_{j=1}^N u_j^2 - \sum_{j=1}^N \sum_{j \neq k} u_j u_k]$
(exercise!)
 - $\Rightarrow Var(\bar{Y}) = \frac{1}{n^2} \frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{N}{N-1} \sum_{j=1}^N (u_j - \mu)^2$
 - $\Rightarrow Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$
- We have found the variance of our estimator
 - How is this different from the formulas you have seen before?

SRS: Properties of $\bar{Y}$

- We have found the variance of our estimator
- Notice the difference between this and the “usual” variance of the sample mean – there is a factor of $\left(\frac{N-n}{N-1}\right) = 1 - \frac{n-1}{N-1}$
- This is an example of a finite population correction, or a **f.p.c.**
 - We will also see the f.p.c. $\frac{N-n}{N} = 1 - \frac{n}{N}$ appear more commonly in our equations, which uses the **sampling fraction** $f = \frac{n}{N}$

SRS: Properties of $\bar{Y}$

- Unfortunately, most of the time **we do not know parameter σ**
- Can we estimate this as well?
- Let us use the **sample variance**
 - $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$
 - Denote by s^2 the **observed value** of $\hat{\sigma}^2$, i.e. $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$
- Is this an unbiased estimator for σ^2 ?

SRS: Properties of $\bar{Y}$

- By repeating a similar calculation to showing $Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$, we can show
 - $E[\hat{\sigma}^2] = \frac{N}{N-1} \sigma^2 = \frac{1}{1-\frac{1}{N}} \sigma^2$
 - (See Problem Sheet 1)
- Hence, $\hat{\sigma}^2$ is a biased estimator of σ^2
 - But $\left(1 - \frac{1}{N}\right) \hat{\sigma}^2$ is an unbiased estimator for σ^2

SRS: Properties of $\bar{Y}$

- Since $Var(\bar{Y}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)$, we see that
 - $\frac{1}{n} \frac{N-n}{N-1} \frac{N-1}{N} \hat{\sigma}^2 = \left(1 - \frac{n}{N} \right) \frac{1}{n} \hat{\sigma}^2$ is an unbiased estimator for $Var(\bar{Y})$
- As this quantity will be used rather often, we will give it its own notation:
 - $\widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N} \right) \frac{\hat{\sigma}^2}{n}$
 - Notice the appearance of the f.p.c.
- What happens as the population size grows large?

SRS: Confidence Intervals for μ

- Yes, sometimes, according to the central limit theorem (CLT) for simple random sampling without replacement from finite populations Hájek (1960)
 - If certain technical conditions hold (!) and if n , N and $N - n$ are all sufficiently large,
 - then $\frac{\bar{Y} - \mu}{\sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}}$ is approximately $N(0,1)$ and
 - $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}$ is a large-sample $100(1 - \alpha)\%$ C.I.

SRS: Confidence Intervals for μ

- But usually we don't know the value of σ
 - Since $\frac{N-1}{N} \hat{\sigma}^2$ is an unbiased estimator for σ^2 , it is common to **replace σ** in the formula with $\sqrt{\frac{N-1}{N}} \hat{\sigma}$
 - And the C.I. becomes $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{\hat{\sigma}}{\sqrt{n}}$.
 - Note this is a random variable. The observed value is $\bar{y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{s}{\sqrt{n}}$.

SRS: Confidence Intervals for μ

- But hang on, that assumes

- $\frac{\bar{Y} - \mu}{\sqrt{\frac{N-n}{N} \left(\frac{\hat{\sigma}^2}{n} \right)}} = \frac{\bar{Y} - \mu}{\sqrt{\widehat{Var}(\bar{Y})}} \sim N(0,1)$, which isn't true...

- However, we do know that if the $Y_1, Y_2, \dots, Y_n$ are i.i.d. normal, then

- $\frac{\bar{Y} - \mu}{\sqrt{\widehat{Var}(\bar{Y})}} \sim t_{n-1}$ distribution

- Therefore, in practice we replace the $z_{\frac{\alpha}{2}}$ percentile with a $t_{n-1, \frac{\alpha}{2}}$ percentile, so the CI becomes

- $\bar{y} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N} \frac{s}{\sqrt{n}}}$

SRS: Sample sizes for μ

- We will say that our goal is to have a C.I. of a certain width, $2d$.
 - What sample size do I need to achieve this width?
- If $Var(\bar{Y})$ is known, the width of the $100(1 - \alpha)\%$ C.I. is
 - $2\sqrt{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} Z_{1-\frac{\alpha}{2}}$
- Hence the appropriate sample size is the n which solves
 - $\sqrt{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} Z_{1-\frac{\alpha}{2}} = d$
- Some calculation shows
 - $n = \frac{1}{\frac{1}{N} + \frac{d^2}{\sigma^2 Z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)}$

SRS: Sample sizes for μ

- The problem with this is... we usually don't know σ . How can we estimate it?
 - **Estimate it:** use the sample variance from your pilot survey (assuming you did one...)
 - **Copy it:** use some **historical data (existing studies)** that exists on the topic and plug-in values found there
 - **Guess it:** if you have a reasonable hypothesis for the distribution of the data, that can give you some information regarding the variance.
 - E.g. you believe the population is normal and you have an idea for the range of the data. Then you could estimate σ by $\frac{Range}{4}$ or $\frac{Range}{6}$ because 95% and 99.7% of values from a normal distribution are within 2 and 3 standard deviations of the mean respectively.

SRS: Sample sizes for μ

- Say **pre-sample estimate for σ is v**
- Then we may use this formula to find an appropriate sample size for us

- $$n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 z^2 \left(1 - \frac{1}{N}\right) \left(1 - \frac{\alpha}{2}\right)}}$$

SRS: Sample sizes for μ

- Then to find the appropriate sample size we need to solve

- $\sqrt{\left(1 - \frac{n}{N}\right) \frac{v^2}{n}} t_{n-1, 1-\frac{\alpha}{2}} = d$ for n

- i.e. to solve $n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$

- The difficulty here is that n appears on both sides of the equation

Summary: SRS for $\mu : \bar{Y}$

- Estimator: $\bar{Y} := \frac{1}{n} \sum_{i=1}^n Y_i$ or $\bar{Y} = \frac{1}{n} \sum_{j=1}^N u_j Z_j$
- $E(\bar{Y}) = \frac{1}{n} \sum_{j=1}^N u_j \frac{n}{N} = \frac{1}{N} \sum_{j=1}^N u_j = \mu$
- $Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2 \Rightarrow \widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}$
- C.I.: $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N-1} \frac{\sigma}{\sqrt{n}}} \Rightarrow \bar{y} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N} \frac{s}{\sqrt{n}}}$
- Sample size: $n = \frac{1}{\frac{1}{N} + \frac{d^2}{\sigma^2 z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)} \Rightarrow n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$

SRS for τ : $\hat{\tau}$

- Estimator: $\hat{\tau} = N\bar{Y}$
- $E(\hat{\tau}) = \tau$
- $Var(\hat{\tau}) = \frac{N-n}{N-1} \frac{N^2}{n} \sigma^2 \Rightarrow \widehat{Var}(\hat{\tau}) = N^2 \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}$
- C.I.: $\hat{\tau} \pm \sqrt{N^2 \frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} z_{1-\frac{\alpha}{2}}$
 $\Rightarrow \hat{\tau} \pm \sqrt{N^2 \left(1 - \frac{n}{N}\right) \frac{s^2}{n}} t_{n-1, 1-\frac{\alpha}{2}}$
- Sample size: $n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 \sigma^2 z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)} \Rightarrow n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$

SRS for $p : \hat{p}$

- Estimator: $\hat{p} = \frac{1}{n} \sum_{i=1}^n Y_i$
- $E(\hat{p}) = p$
- $Var(\hat{p}) = \frac{N-n}{N-1} \frac{1}{n} p(1-p) \Rightarrow$

$$\widehat{Var}(\hat{p}) = \left(1 - \frac{n}{N}\right) \frac{1}{n-1} \hat{p}(1 - \hat{p})$$
- C.I.: $\hat{p} \pm \sqrt{\left(1 - \frac{n}{N}\right) \frac{1}{n-1} \hat{p}(1 - \hat{p})} t_{n-1, 1-\frac{\alpha}{2}}$
- Sample size: $n = \frac{1}{\frac{1}{N} + \frac{d^2}{p(1-p) z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)}$

Stratified Random Sampling for $\tau: \hat{\tau}_{st}$

- Estimator: $\hat{\tau}_{st} = \sum_{i=1}^L N_i \bar{Y}_i$
- $E(\hat{\tau}_{st}) = \tau$
- $Var(\hat{\tau}_{st}) = \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$
 $\Rightarrow \widehat{Var}(\hat{\tau}) = \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i} \right) \frac{\hat{\sigma}_i^2}{n_i}$
- C.I.: $\hat{\tau}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})} \Rightarrow \hat{\tau}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})}$
 $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{n_h}$
- Sample size: $n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{\frac{d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{N_i - 1}}$

Stratified Random Sampling for $\mu : \bar{Y}_{st}$

- Estimator: $\bar{Y}_{st} := \frac{1}{N} \hat{t}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \bar{Y}_i$
- $E(\bar{Y}_{st}) = \mu$
- $Var(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$
 $\Rightarrow \widehat{Var}(\bar{Y}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i} \right) \frac{\hat{\sigma}_i^2}{n_i}$
- C.I.: $\bar{y}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})} \Rightarrow \bar{y}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$

$$df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}} \text{ where } k_h = \frac{N_h(N_h - n_h)}{N^2 n_h}$$

- Sample size: $n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{\frac{N^2 d^2}{z^2 \frac{1-\alpha}{2}} + \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{N_i - 1}}$

Stratified Random Sampling for $p : \hat{p}_{st}$

- Estimator: $\hat{p}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \hat{p}_i$
- $E(\hat{p}_{st}) = p$
- $Var(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{N_i - n_i}{N_i - 1} \frac{1}{n_i} p_i (1 - p_i) \Rightarrow$

$$\widehat{Var}(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i}\right) \frac{1}{n_i - 1} \hat{p}_i (1 - \hat{p}_i)$$

- C.I.: $\hat{p}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{p}_{st})} \Rightarrow \hat{p}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{p}_{st})}$

$$df \approx \frac{\left(\sum_{h=1}^L k_h s_h^2\right)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}} \text{ where } k_h = \frac{N_h(N_h - n_h)}{N^2 n_h}$$

- Sample size : $n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{p_i(1-p_i)}{N_i - 1}}{\frac{N^2 d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{p_i(1-p_i)}{N_i - 1}}$

Stratified Random Sampling :Sample Allocation

- Choose sample sizes $n_1, \dots, n_L$ that are in the same proportion to the total sample size n as the $N_1, \dots, N_L$ are to the total population size N
- Therefore $n_1 = \frac{nN_1}{N}, n_2 = \frac{nN_2}{N}, \dots, n_L = \frac{nN_L}{N}$
 - And $a_1 = \frac{N_1}{N}, a_2 = \frac{N_2}{N}, \dots, a_L = \frac{N_L}{N}$

Neyman Allocation: optimal under sample size constrain

- Optimization problem:

– minimize the function

$$f(n_1, \dots, n_L) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$

– over all $n_1, n_2, \dots, n_L$

– with the constraint $g(n_1, \dots, n_L) = n - \sum_{i=1}^L n_i = 0$

- Solution:

$$n_i = n \frac{\sqrt{\frac{N_i}{N_i - 1}} N_i \sigma_i}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i - 1}} N_i \sigma_i} \text{ in terms of the corrected pop.}$$

var.

Optimal Allocation under cost constrain

- Optimization problem:

– minimize the function

$$f(n_1, \dots, n_L) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$

– over all $n_1, n_2, \dots, n_L$

– with the constraint $g(n_1, \dots, n_L) = c_0 + c_1 n_1 + c_2 n_2 + \dots + c_L n_L - c = 0$

- Solution:

$$n_i = \frac{(c - c_0) \left(\sqrt{\frac{N_i}{N_i - 1}} \frac{N_i \sigma_i}{\sqrt{c_i}} \right)}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i - 1}} N_i \sigma_i \sqrt{c_i}}$$

Lagrange Multipliers Method

- Consider the problem
 - Maximize/minimize $f(x_1, \dots, x_n)$
 - Subject to the constraint $g(x_1, \dots, x_n) = 0$
- We solve this by
 1. Writing down the **Lagrangian**
$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) - \lambda g(x_1, \dots, x_n)$$
 2. Find the **stationary points** of $L(x_1, \dots, x_n, \lambda)$, i.e.
where $\frac{\partial L}{\partial x_i} = 0$ for $i = 1, \dots, n$ and $\frac{\partial L}{\partial \lambda} = 0$
 3. Evaluate $f(x_1, \dots, x_n)$ at each stationary point,
determine which is the maximum/minimum

Cluster Sampling for τ : $\hat{\tau}$

- Estimator: $\hat{\tau} = N \frac{1}{n} \sum_{i=1}^n Y_i$, where $Y_i = \sum_{j=1}^{M_i} y_{ij}$ is total of observations in the i -th cluster
- $E(\hat{\tau}) = \tau$
- $\widehat{Var}(\hat{\tau}) = N^2 \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}_c^2}{n}$, where $\hat{\sigma}_c^2$ is the sample variance of the cluster totals
- C.I.: $\hat{\tau} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau})}$
- Sample size: $d = z_{1-\frac{\alpha}{2}} \sqrt{Var(\hat{\tau})}$ for n

Two-stage cluster Sampling for τ : $\hat{\tau}$

- Estimator: $\hat{\tau} = N \frac{1}{n} \sum_{i=1}^n \hat{Y}_i$, where $\hat{Y}_i = M_i \frac{1}{m_i} \sum_{j=1}^{m_i} Y_{ij}$ is SRS unbiased estimator of the i -th cluster total
- $E(\hat{\tau}) = \tau$
- $\widehat{Var}(\hat{\tau}) = N(N - n) \frac{1}{n} \hat{\sigma}_c^2 + \frac{N}{n} \sum_{i=1}^n M_i(M_i - m_i) \frac{1}{m_i} \hat{\sigma}_i^2$,
where $\hat{\sigma}_c^2$ is the sample variance of the estimated cluster totals and $\hat{\sigma}_i^2$ is the sample variance inside cluster i
- C.I.: $\hat{\tau} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau})}$
- Sample size: not required